AVTools System Designer — User Guide

Version 3.1.0 · Revit 2025, 2026 and 2027 · J. Stevens BIM Acoustics, LLC

This guide walks through every tab of **AVTools System Designer** (Revit 2025, 2026 and 2027) — what it does, how to drive it, and what to expect at each step. Read it cover-to-cover for the full workflow, or jump to the section that matches your current task. Room Exchange and SpecTool have guides of their own.

What's New in v3.1.0

- **Revit 2027.** One installer adds System Designer, SpecTool and Room Exchange to Revit 2025, 2026 and 2027.
- **F1 help.** Hover over any AVTools ribbon button and press F1 to open its help page on bimacoustics.net.
- **Ribbon.** The BIM Acoustics tab has two panels: **AV Tools** (System Designer, SpecTool) and **Utilities** (Room Exchange, Lucius, License Manager). The empty Acoustics Tools panel is gone.
- **Lucius transcript opt-in.** Anonymous transcript capture is off unless you turn it on. The first time you open Lucius you are asked once; you can change it later with the **Share anonymous transcripts** tick box in the Lucius window or on the About tab, which stay in step.
- **Iso-map.** Coverage is sampled on a grid no coarser than a quarter of the loudspeaker spacing, so each loudspeaker gets its own contour instead of contours that merge and split. Drawing a room's map in a second view no longer stops with "type name JSBA_DFM_… is already in use".
- **Messages** from System Designer now open in front of its window instead of behind it.
- **Tag families:** the "JSBA Circuit ID Tag w Tap" family is no longer installed; use "JSBA Circuit ID Tag w Watts".
- **Room Exchange:** an **Include loudspeakers** tick box, and EASE 5 loudspeaker targets that lie flat just under the ceiling. See the Room Exchange User Guide.
- **Security updates** to the bundled components, made for the Autodesk App Store review.

From v3.0.0

- **Room Exchange joins the AVTools ribbon (Pro).** It turns Revit rooms into an EASE 5 room model (a DWG of closed faces, one layer per surface material), and brings loudspeakers from an EASE project back into Revit. See the separate Room Exchange User Guide.

- **Room Acoustics finds surfaces with the Room Exchange engine.** Each room is read as a closed shell, with the same capture and fill as an EASE export: floors, ceilings, walls, glass, doors, columns and operable partitions, which way each faces, and the air volume, from the shell rather than from Revit's room boundaries. The material list is unchanged. A room the engine cannot close falls back to the earlier Revit-boundary method, with a note.
- **Assign Materials** lists one row per kind of surface material (not per Revit type), and has a **Leave out** column for scenery and structure that should not count in the RT60.
- **Refresh** on Room Acoustics reads the listed rooms from the model again and recalculates. A read that was slow because of one or two very heavy elements offers to leave them out (see Room Acoustics).
- **Reset All Design State** now also removes what AVTools placed in the model, after showing what it will remove and asking first.
- **Placement:** a loudspeaker hosted on a linked ceiling gets the room's level in Schedule Level when Revit leaves it empty.
- **Layout:** the grid-rotation detector no longer turns a room it reads as square to the axes onto a single skewed wall.
- **Lucius privacy:** the model's name, linked models' names, folder paths and file names are replaced with short aliases before anything is sent to the assistant.
- **Installing v3.0.0** replaces v2.3.x and the 2.4.0 beta builds. Project state, license activation and EULA acceptance carry forward.

From v2.3.3 – v2.3.5

- **The free trial unlocks every Pro feature.** For 14 days you can use everything in Pro: circuiting, amps & cabling, coordination, SpecTool, Room Exchange and the Lucius tools. After the trial, subscribe to Standard or Pro and click **Validate Now** on the About tab.
- **Changing the license email works cleanly.** Enter a different email on the About tab and click Validate Now: that email is checked on its own. If the license server turns a license down, the About tab says so and shows the server's reason.
- **Encrypted connections.** License activation and registration, the Lucius assistant and SpecTool all connect over HTTPS.
- **Remove Selected Rooms** cleans up what AVTools placed for the removed rooms (see the Rooms tab).
- **Circuit schedule** lists loudspeakers only and keeps your own column, sort and filter edits when you update it (see Circuiting).
- **Same-level rack assignment** option in Amps & Cabling.
- **Manual loudspeaker pick** warns before using a family type that doesn't look like a loudspeaker or has no coverage data.
- **Name in Windows and Revit:** Apps & Features and Revit's Signed Add-In prompt show *AVTools System Designer* by J. Stevens BIM Acoustics LLC. On first launch, choose **Always Load**.

- **Internet connection:** Lucius and SpecTool need an internet connection. Everything else in System Designer, and Room Exchange, works offline once your license has been activated.

What This Product Is

AV Tools Suite — AV Systems System Designer is a Revit add-in that provides a complete workflow for designing, analyzing, and documenting distributed ceiling loudspeaker systems. It runs as a dockable panel inside Revit 2025, 2026 and 2027 and supports both host-model and linked-model architectural coordination.

The product has two paid editions plus a free legacy version:

- **Free (v1.2.1)** — Revit 2022–2024. Room selection, speaker layout, placement, direct-field coverage, Lucius chat (guidance only).
- **Standard (v3.1.0)** — Revit 2025, 2026 and 2027. Adds room acoustics (RT60, STI, material assignment), Lucius AI chat with read tools.
- **Pro (v3.1.0)** — Revit 2025, 2026 and 2027. Adds circuiting with zones, amps & cabling, coordination/clash detection, advanced rack workflow, Lucius AI write tools (currently material assignment), **SpecTool** (CSI Part 2 spec generator) and **Room Exchange** (Revit rooms to EASE 5, EASE loudspeakers back to Revit). SpecTool and Room Exchange have their own guides.

Pricing: Standard $60/month or $600/year. Pro $99/month or $990/year. 14-day free trial that unlocks every Pro feature.

The Workflow (Eight Tabs)

The tool is organized as a step-by-step workflow. Tabs in order:

1. **Rooms** — pick which rooms are in design scope. Always first. Includes the Add Temp Room workflow for spaces not yet laid out cleanly in the architectural model.
2. **Configuration** — set host mode, spacing, listener height, layout, speaker selection per room. Run **Refresh Calculations** then **Commit Placement** here.
3. **Direct-Field Coverage** — compute SPL uniformity from placed speakers. Run **Compute Results**, optionally **Draw Iso-Map**.
4. **Room Acoustics** — surfaces found with the Room Exchange engine; assign acoustic materials, then calculate RT60 (Sabine, Norris-Eyring, Arau-Puchades) and estimated STI.
5. **Circuiting** *(Pro)* — assign circuit IDs, zones, taps, modes (70V/100V/Low-Z); generates the JSBA Loudspeaker Circuit Schedule.
6. **Amps & Cabling** *(Pro)* — rack discovery, nearest-rack assignment, wire gauge, line loss, wiring diagrams.
7. **Coordination** *(Pro)* — clash detection between AV devices and MEP/lighting/structural elements.

8. **About** — version, licensing, transcript opt-in, support tools.

Loudspeakers are placed in the host model. Rooms can come from the host or linked architectural models. Works from either Floor Plan or Reflected Ceiling Plan views.

The BIM Acoustics ribbon tab has two panels. **AV Tools** holds **System Designer** (this panel) and **SpecTool**; **Utilities** holds **Room Exchange**, **Lucius** and **License Manager**. Hover over a button and press F1 for its help page.

Rooms Tab — Source Column and Unified Link Scan

The Rooms tab is the always-first entry point. Every session starts here.

Single unified scan (new in v2.2): The previous "Host vs Linked" source dropdown has been retired. The Rooms tab now scans the host document plus every loaded Revit link in a single pass, and a new **Source** column shows which file each room came from. Rooms from unloaded links don't appear (link workset state is honored), and a duplicate-room dedup pass cleans up the case where a container model nests the same architectural link that's also loaded standalone.

Practical effect: for most projects you click **Rescan Rooms** once and see everything. No more switching modes when a project has rooms in multiple linked files.

Remove Selected Rooms takes the checked rooms out of design scope and cleans up what AVTools placed for them: it deletes the loudspeakers AVTools placed in those rooms, their iso-maps, wiring, homerun notes and circuit tags, clears circuit and cabling parameters on racks and other equipment that referenced them, and resets the rooms to "Not Started". Loudspeakers you placed by hand (without AVTools stamps) are not touched. The completion dialog reports how many elements were removed, and Revit Undo reverses it. Rooms that contain only racks or other AV equipment stay "Not Started" — only loudspeakers count toward "Speakers Placed".

Reset All Design State (new in v3.0.0) does the same for the whole project: it first counts what it would remove (loudspeakers and equipment AVTools placed, iso-map items, wiring and homerun notes, circuit tags and other design lines) and asks before anything is deleted. *Yes* removes them by the same rules as Remove Selected Rooms and resets the saved design state on every tab.

Rooms Tab — Add Temp Room *(new in v2.2)*

Real projects regularly have rooms that aren't bounded in the architect's Revit model — large prefunction halls, open food halls, exterior plaza spaces, or rooms still being modeled by the architect when AV layout is happening. **Add Temp Room** is a single Rooms-tab workflow that handles every variant:

- **The space has no Revit Room at all.** Pick a number, a name, and a ceiling height; trace the perimeter in the active plan view. AVTools surfaces a synthetic **Temp Room** row in the Rooms tab. All downstream features (Configuration, Direct-Field Coverage, Circuiting, Amps & Cabling, Draw Wiring) treat it like a normal room.
- **The Revit Room exists but is unbounded (Area = 0).** Same workflow — pick the row, trace the perimeter. AVTools creates a stamped Filled Region tied to that room number that the next room scan picks up as a layout boundary.

The Filled Region is an annotation — it does **not** propagate through Revit links, so other disciplines linking the AV model never see it.

Temp Room → Real Room auto-association: when speakers placed in a Temp Room polygon overlap a real Revit room (a corridor, a room with a bad centroid, etc.), AVTools auto-resolves on the next **Rescan Rooms**. Speakers are re-stamped to the real room's number, the real room's row inherits the temp's design state (Speakers Placed / Circuiting Complete / Cabling Complete — promote-only, never demote), and the Temp Room row is hidden. If the temp polygon spans multiple real rooms, AVTools shows a warning dialog and waits for the polygon to be refined. If the polygon sits in genuinely unbounded space, the temp row stays visible as a normal scaffold.

Temp rooms are excluded from **Room Acoustics** because they're documentation-only placement scaffolds. If you want acoustic analysis on the underlying space, the real Revit room must exist and be in scope.

Rooms Tab — Room-Type Heuristic

When you add a room to scope (Use checkbox + Next), AVTools matches the room's name against two keyword lists and seeds Configuration-tab fields accordingly:

Category	Name tokens (any match, case-insensitive)	Seeded values
Presentation	classroom, instruction, training, lecture, meeting, conference, board room, ballroom, auditorium, theater, lab, library, cafeteria, cafe, restaurant, dining, food court, lounge, break room, bar, pub, tavern, bistro, club	Spacing **Minimum Overlap** · Listener **4.0 ft** (seated) · Target SPL **82 dB**
Circulation	lobby, corridor, hallway, hall, vestibule, concourse, restroom, toilet, entry, foyer, atrium	Spacing **Edge to Edge** · Listener **6.0 ft** (standing) · Target SPL **76 dB**
Fallback (name doesn't match either list)	—	Spacing & Listener stay at the project's **Global Defaults** · Target SPL **82 dB**

The heuristic only fires when a row is **still at the project's Global Defaults** for spacing and listener height. If you've manually edited those fields on a row, the heuristic backs off and won't overwrite your work on a

re-add. These are keyword presets you can override — not magic, not AI-driven, just a starting-point convenience.

Speaker Layout & Placement

Three spacing modes:

- **Center-to-Center** — speaker centers on a regular grid
- **Edge-to-Edge** — coverage circles touch
- **Minimum Overlap** — coverage circles overlap to a defined percentage

The tool derives coverage radius from ceiling height, listener height, and the speaker's coverage angle, then fills the room with a grid. Spacing is **deterministic**: identical inputs in the same model context produce repeatable outputs.

Host modes:

- **Ceiling / Auto** — speaker hosted to the detected ceiling
- **Reference Plane** — for rooms without standard ceilings (exhibit halls, catwalks, open-to-deck spaces). v2.2 uses a place-on-lower-side strategy that consistently brings speakers down cone-toward-listener on horizontal reference planes (the common pattern for tall ballrooms and exhibit halls).
- **Fallback Z** — fixed elevation when neither ceiling nor reference plane is appropriate

Tile centering is supported for ACT ceiling grids (2×2, 2×4) so speakers align to tiles rather than landing on grid lines.

Linked model rooms: rooms from the architect's linked model are fully supported. Speakers are placed in the host AV model. Copy/Monitor is not required. A loudspeaker hosted on a face of a linked ceiling gets no level from Revit; AVTools writes the room's level to its **Schedule Level** at placement (only when Revit left it empty), so schedules and Elevation from Level read correctly.

Acoustic data files for speakers — only `.spk` (EASE SPK format, with optional `.lob` / `.phs` / `.fed` / `.frd` / `.fvt` siblings) and `.clf` (Common Loudspeaker Format, CLF1 + CLF2) are supported. `.gll`, manufacturer proprietary binary formats, and `.cf2` as balloon data are **not** supported. If a manufacturer only provides `.gll`, request CLF data from them; without supported data, the family falls back to default sensitivity (78 dB at 1W/1m) and approximate Q from manual family parameters. A manual loudspeaker pick warns before using a family type that doesn't look like a loudspeaker or has no coverage data.

Nudge tool: for fine-tuning speaker positions after auto-layout. Click Nudge on a Configuration row → AVTools selects all that room's placed speakers in Revit's UI and hands focus to the canvas → use Revit's native arrow-key nudge with Shift/Ctrl modifiers → Esc when done.

Grid θ rotation: blank = auto-detect from the room polygon's walls. A room that reads as square to the axes keeps its grid square; the longest-edge fallback is used only when the walls give no clear direction.

Override per-row when auto-detect picks the wrong axis (irregular rooms, ceiling tile grid running off-axis from the room polygon, soffit-driven orientation).

Direct-Field Coverage Analysis

Compute Results scans placed speakers and computes per-room uniformity using each room's Target SPL and Listener Height. Results display:

- Speakers Placed count
- Target SPL
- Min / Max SPL
- Range
- Uniformity bands: **% within ±1 dB**, **% within ±2 dB**, **% within ±3 dB**, **% > +3 dB**

Yellow cells flag metrics outside preferred thresholds; review by adjusting spacing, layout, target SPL, or speaker count, then re-Commit Placement and re-Compute Results.

Draw Iso-Map draws bucketed iso-line contours on the active view at the listener plane. **Clear Map** removes them. The lines are drawn 2, 4 and 6 dB below the room's 95th-percentile level, which on an even grid of loudspeakers sits close to the level under each loudspeaker. Coverage is sampled on a grid no coarser than a quarter of the loudspeaker spacing (1 ft at the finest), so each loudspeaker gets its own contour.

Coverage map line styles (new in v2.2): the iso-coverage contour lines use Line subcategories with built-in colors (green / yellow / red) instead of per-element view overrides, so coverage maps render in color on the parent floor plan **and** on every dependent plan view placed on sheets.

The coverage engine models **direct-field only** — no reflections, no full-room simulation. Reflections, RT60, and STI are computed separately by the Room Acoustics tab.

If results show `-999` for a room, that room has speakers without usable tap/wattage/sensitivity data: verify Loudspeaker family + acoustic data file (`.spk` / `.clf`) are set, Tap Watts is non-zero, then re-compute.

Room Acoustics

How surfaces are found (new in v3.0.0). Each listed room is read with the **Room Exchange engine**, the same capture and fill as an EASE export from Room Exchange. The engine fills the room's air and finds every surface that air touches, so the RT60 is given a closed shell: the floor, every ceiling (ACT, gypsum, clouds), walls, glass, doors, columns and operable partitions, each with the way it faces, and the air volume of the shell. A room's own volume override still wins over the shell volume. If the engine cannot close a room, that room falls back to the earlier Revit-boundary method and the tab says so.

- **Reading takes time.** The rooms of a level are read once per session; Assign Materials and recalculations reuse that read. A first read of a large linked model can take minutes.
- **Refresh** reads the listed rooms from the model again and recalculates. Use it whenever anything in the rooms has changed.
- **Slow reads.** After a read of more than about 90 seconds in which one or two very heavy scenery or structure elements took most of the time, a window lists them with **Show in Revit** for each, plus **Leave out and read again** and **Keep it**. Revit stays usable while it is open and the results show behind it. An element left out that stands inside a room comes back as a gray row in Assign Materials, where you can put it back.
- **Rooms on several levels.** Rooms that span multiple levels (a theater's house, stage and catwalk drawn as separate rooms on different levels) are read one level at a time here. To treat them as one space, export them together with Room Exchange.

RT60 calculation — three formulas, eight octave bands each:

- **Sabine** — standard reverb formula, assumes diffuse field
- **Norris-Eyring** — better for highly absorptive rooms
- **Arau-Puchades (AP)** — classical 1988 formula. Per-axis α from surfaces perpendicular to each axis; per-axis RT uses Eyring with that axis's α; final RT is the surface-weighted geometric mean of the three axis RTs. Robust default.
- **Arau-Puchades (Full)** — adds ISO 9613-1 air absorption per band. More accurate at 4 kHz+ where air absorption matters; numerically fragile on lopsided-absorption rooms (skipped at 63 Hz, falls back to Sabine when the geometric mean collapses).

Octave bands computed: **63, 125, 250, 500 Hz, 1, 2, 4, 8 kHz**.

AP > Eyring at 1–2 kHz in typical meeting rooms is **expected** — absorbent ceiling + carpet floor + hard gypsum walls is exactly the lopsided-absorption case AP captures and Eyring's uniform-mixing assumption hides.

STI (Speech Transmission Index) — estimated from RT60 bands, background noise, speaker Q factor, and N factor. Rated per IEC 60268-16:

STI	Rating
0.75–1.00	Excellent
0.60–0.74	Good
0.45–0.59	Fair
0.30–0.44	Poor
0.00–0.29	Bad

Targets: ≥ 0.50 general; ≥ 0.60 for critical-speech spaces; ≥ 0.50 mass-notification minimum.

Critical Distance and D/R Ratio: computed per room. Dc = 0.141 × √(Q × R) where R is the room constant. D/R is computed and displayed alongside.

Operable partition detection: rooms separated by operable partitions are detected. RT60 is computed for the combined state and exportable to CSV.

Ceilings: each kind of ceiling in a room (for example a GYP perimeter and a 2'×6' ACT field) is its own row in Assign Materials, with its own area. A ceiling that spans several rooms is counted in each room only over that room's own air.

Reference curve overlay: the RT60 graph supports a reference curve dropdown with preset target curves (Conference/Meeting, Classroom, Lecture Hall, Theater, Worship, Courtroom, Ballroom, Exhibit Hall) — sourced from ANSI/ASHRAE/Acoustical Society guidance. Manual reference values can also be entered per band, useful for: (1) comparing against another model (EASE, ODEON, CATT), (2) overlaying measured field data to validate prediction, or (3) tuning material assignments on renovation projects until the AVTools prediction matches measurement.

Schedule preservation: running RT60 on one level does not wipe results for rooms at other levels. Calculations persist across runs.

Temp Rooms are not analyzed. Acoustics calculations require a real Revit Room; Temp Rooms (placement scaffolds) are excluded from this tab.

Material Assignment

Click **Assign Materials** to open the picker. Built-in material database covers ~2,500 records sourced from a published German absorption-coefficient study, plus support for user EASE `.mat` folders and custom user materials. Mappings are made at the **family/type level** — assign once, applies everywhere that type appears across the project.

Rows (new in v3.0.0): the engine's surfaces are grouped into one row per kind of surface material — for example one row for all gypsum walls, one for the ACT ceiling, one for the carpet floor — with the Revit families and types behind each row listed beside it. Assignments you made to those types before carry over.

Leave out column (new in v3.0.0): tick it to keep scenery or structure out of the RT60. It is enabled only on rows made wholly of generic models, casework, specialty equipment, furniture, stairs, ramps, steel framing or columns, and never on an operable partition, a seal or a room separation; a dimmed box says why in its tooltip. Elements already left out come back as gray, ticked rows so you can put them back. OK reads the rooms again; if a room that closed before no longer closes, you are told and offered to put the elements back.

Surface Area Diagnostics panel shows per-axis areas (E/W/N/S/F/C) with three pair checks:

- **E/W pair:** OK if matched within 20%

- **N/S pair:** OK if matched within 20%
- **F/C pair:** OK if matched within 25% (more permissive because stacked or partial ceilings can drift a few percent from floor area)

Mismatches don't block calculation but suggest geometry to recheck. A real difference is not always an error: a room with an absorptive treatment on one wall and gypsum on the opposite wall will show it. When a room falls back to the Revit-boundary method, the warnings also name a floor or wall type that was left out of the RT60, or a floor type with no material assigned.

User materials: Add Material (new entry), Add User Folder (bulk import, supports EASE .mat and JSON conforming to schema), Edit (only enabled for user materials), Duplicate (any material), Delete (with confirmation; orphans existing assignments).

Common workflow for missing-band values: find closest library match → Duplicate → fill in missing bands from manufacturer cut sheet or hand-calc → adjust name → Save → Assign.

Circuiting *(Pro)*

Assigns circuit IDs per room or zone. First time on the tab: one circuit per room (e.g., `02.31.05`). **Add Zone** splits a room's circuit into sub-zones (`/A` , `/B` , `/C`), useful for divisible rooms with operable partitions or separate audience/dais areas. The first-click zone split correctly assigns `/A` to the picked speakers and `/B` to the rest.

Modes:

- **70V** — North American constant-voltage; multi-tap transformers in parallel; typical paging/BGM/presentation
- **100V** — same as 70V at 100V line voltage; European standard, also long cable runs in NA
- **Low-Z** — direct low-impedance distribution, no transformer; high-fidelity music, single-speaker zones, theater

Apply to Selected / Apply to All (new in v2.2): explicit-action buttons replace the previous auto-apply-on-dropdown-change for system voltage mode. Picking a new default no longer pre-emptively writes to every row — select rows, then click Apply. The Apply-button inventory is intentional: these buttons exist where they make sense (system voltage on Circuiting, wire gauge on Amps & Cabling, host/layout on Configuration) and don't exist elsewhere.

Tap selection: nearest tap ≥ required power algorithm. Power-overload highlighting: green < ~80% Max Watts, yellow 80–100% (tight, no headroom), red > 100% (must fix before issuing documentation).

Commit Circuiting writes circuit ID, tap setting, and rack ID to every speaker as part of a single commit transaction; the values stick on subsequent reads. **Update Schedule** does a Commit + rebuild rather than a rebuild-only, so changes to wire gauge, tap settings, or line loss show up immediately in the circuit schedule view.

JSBA Loudspeaker Circuit Schedule is a multi-category ViewSchedule with these columns: Circuit, Destination, Type, Tap (W), Qty, Spkr (Ω), Circuit (Ω), AWG, Loss (dB), Measured (Ω). The schedule is tracked by Revit UniqueId, so renaming it on a sheet doesn't break the linkage. It lists loudspeakers only (racks, amplifiers and other AVTools-managed equipment are excluded). Fields, column order, sorting and filters are set only when the schedule is first created; after that, Update Schedule refreshes the data and keeps your own customizations.

Tag families are bundled and auto-load on first project use — JSBA Circuit ID Tag (multi-category, circuit ID only for coordination drawings) and a variant with Tap Watts displayed (for installation drawings, eliminates installer error of default-tap-set).

Amps & Cabling *(Pro)*

Rack workflow:

1. **Rack Locations** — scans the entire project for rack candidates (across host + linked models) using keyword matches on family names: rack, cabinet, enclosure, Middle Atlantic product codes. Family names with underscores (e.g., `MAP-SR_Series`, `SR_Series_Wall_Rack`) are normalized to spaces before matching so the SR series is recognized correctly.
2. **Rack Family Selection dialog (new in v2.2)** — the rack scan can pull in hundreds of false-positive matches on projects with furniture, IT racks, network cabinets, or other "rack/cabinet/enclosure" families that aren't AV equipment racks. After the scan, a selection dialog lists every keyword-matched family with its instance count and an example room, plus a tick-box to include or exclude. Your choice is **persisted at user scope** (in `%APPDATA%\BimAcoustics\UserState.json`), so the rack selection follows you across every project you open on your machine. New families appearing in subsequent scans are flagged with a **"New"** status badge so they can be reviewed without disrupting existing selections.
3. **Pick Racks in Model** — fallback when the category-based scan misses racks (e.g., racks modeled in Generic Models). User picks racks manually in Revit selection mode.
4. **Assign Rack IDs** — writes the rack ID parameter per a configurable naming convention.

Nearest-rack assignment routes each circuit to its nearest rack by room-centroid → rack-location distance (TSP nearest-neighbor). Manual override per circuit available via dropdown — useful for keeping all circuits in one room on the same rack, AV/IT separation, or dedicated VIP-room racks.

Only assign racks on the same level: when this box is checked, a room with no rack on its own level is left unassigned instead of being matched to a rack on another floor. Unchecked (the default) keeps the fallback to other floors. Rooms and racks with no Revit level association get their level from their numbering (e.g. room `02.34.00`, rack `ER-02.01.01` → Level 02).

Bulk rack assignment (new in v2.2): multi-select rows in the Amps grid, choose a rack from the right-side panel, click Apply. Every selected row updates to that rack in one click — useful when the auto-nearest-rack assignment doesn't match your design intent and you want to force a block of circuits to a

specific equipment room. **Recompute Nearest Racks** re-runs auto-assignment and is idempotent (Ctrl+Z to revert).

Apply to Selected / Apply to All (new in v2.2): explicit-action buttons replace auto-apply-on-dropdown-change for wire gauge. Pick a new default, then choose how to apply it. Per-row overrides remain available.

Local Amp flag flips a circuit from a centralized rack to an in-room amp; cable-run estimation drops dramatically.

Estimated Longest Run is a Manhattan-distance approximation for line-loss math only. It is **not** a cable-quantity takeoff — it doesn't account for vertical drops, tray routing, conduit penetrations, service loops, pull boxes, or rack-side dressing. For takeoff, use actual routed cable paths from cable tray drawings plus 10–20% spare.

Line loss thresholds: green < 0.4 dB, yellow 0.4–0.75 dB, red > 0.75 dB. **Damping factor (Low-Z only):** green > 20, yellow 10–20, red < 10.

Wiring styles: Chamfered (45°/90° clean angles, default) or Arc (smooth curves). **Home Run Notes** are native Revit text annotations defaulting to the rack ID (e.g. "TO ER-02.39.16") and editable per-row from the AVTools panel so they persist across redraws.

Coordination *(Pro)*

Detects clashes between placed loudspeakers and other elements (lighting, mechanical, sprinklers, structural framing, cable trays, conduits) across host + linked MEP, structural, and lighting models.

Sensitivity presets:

- Physical (0") — geometric overlaps only
- Tight (0.25–2") — default
- Normal (0.5–4")
- Conservative (1–5")

Lighting specifically uses 0" / 3" / 6" / 12" tiers per industry convention.

Issue tracking: each conflict has a status — Open (red, default), Coordinate (yellow, flagged for cross-discipline), Revisit (parking lot), or Resolved (green/hidden). Status persists across refreshes; markers are matched to current conflicts by speaker ID + conflict ID. Bundled Coordination Issue Marker family auto-loads on first refresh.

Heads-up: the engine detects clashes against Revit family elements in the listed disciplines. It does **not** detect 2D detail items, drafting components drawn as symbols, or imported DWG underlays.

Room Exchange *(Pro)*

Room Exchange is the third button on the AVTools ribbon. It exports the rooms you tick as an **EASE 5** room model: Room Exchange fills the rooms' air, finds every surface the air touches, names each face by the material on the room side, and writes a DWG of closed faces with one layer per material, which EASE imports as a closed volume with one Selection Set per layer. A DXF choice is offered for other programs. Its **Import Loudspeakers** tab places the loudspeakers of an EASE project back into the Revit model, matched to your loudspeaker families and aimed with the EASE angles.

Room Exchange uses the same engine that finds the surfaces on the Room Acoustics tab. It is included in Pro and runs during the free trial. Everything about choosing rooms, the origin marker, the checklist and reading the result in EASE is in the separate **Room Exchange User Guide**.

Lucius AI Assistant (Inside the Add-In)

Lucius is embedded inside Revit as a chat panel. Powered by Claude (Anthropic). Voice input via push-to-talk (Whisper STT) plus text. Lucius reads your live project state via tool use rather than guessing. It also knows Room Exchange and can answer questions about the last export.

What Lucius can read in Standard + Pro:

- Project info (the model is named by a short alias)
- Rooms in AV scope or all rooms
- Per-room metadata
- Placed speakers per room
- Rooms needing attention (unmapped materials, coverage gaps)
- Surface-to-material mappings
- Acoustic material database search

What Lucius can write (Pro only):

- Assign an acoustic material to a surface (with persistence)

What Lucius does NOT do:

- Place speakers — that's the user's job via the Configuration tab
- Commit circuits, run RT60 or coverage calculations, modify the model beyond the one material write
- Read circuit/cabling/coordination/RT60 result data programmatically — Lucius can explain the math, the workflow, and the column meanings, but the user runs the calculations and reads the results from the UI

What is sent: before anything goes to the assistant, the model's name and the names of linked models are replaced with short aliases (computed on your computer), and folder paths and file names are removed. Room numbers and names are sent as they are.

Transcripts: anonymous capture of Lucius conversations, to help improve Lucius, is off unless you turn it on. You are asked once, the first time you open Lucius (No is the default). Change it any time with the **Share anonymous transcripts** tick box in the Lucius window or the transcript tick box on the About tab; the two stay in step. No project geometry or identifying information is captured.

This is intentional: AVTools is **deterministic**, and Lucius guides the user through running the deterministic tool. Lucius does not introduce randomization or "AI placement decisions."

What This Product Is NOT

- **Not a full acoustic simulation tool.** No ray tracing, no diffraction, no room modes. Compare to EASE/AFMG/Treble for full acoustic simulation; AVTools is a faster, BIM-integrated first-order design tool used by AV consultants. Room Exchange is the bridge when a room needs EASE.
- **Not a replacement for engineering judgment.** First-order design assistance, not a guarantee of code compliance or measured performance.
- **Not a substitute for measured field data** on critical-speech or life-safety spaces.

It **does** produce repeatable, BIM-integrated, documentation-ready loudspeaker designs significantly faster than the manual Revit + spreadsheet + standalone-acoustics workflow most AV consultants use today.

Common Questions

Q: What projects is this designed for?
Multi-room distributed loudspeaker systems — convention centers, corporate offices, hospitality, education, healthcare, transportation hubs, houses of worship. Anywhere with many rooms needing consistent ceiling speaker coverage and (in Pro) circuiting and acoustic analysis to support documentation.

Q: Does it work with linked models?
Yes — this is the primary workflow. Rooms typically come from the architect's linked model; speakers are placed in the host AV model. The Rooms tab scans the host doc plus every loaded link in a single pass, and a Source column shows which file each room came from.

Q: Which Revit versions are supported?
v3.1.0 (current paid release): Revit 2025, 2026 and 2027. v1.2.1 (free legacy): Revit 2022, 2023, 2024.

Q: What changed in Room Acoustics in v3.0.0?
Surfaces now come from the Room Exchange engine: a closed shell per room, read from the model's geometry rather than from Revit's room boundaries. Rooms where the earlier method dropped the floor, part of a ceiling or a treated wall (common with linked models) now get them, so their RT60 comes out shorter than before. Assign Materials shows one row per kind of material; assignments made before carry over. Reading takes longer than before, once per session; use Refresh after changing the model.

Q: How do I get a room into EASE?
Use **Room Exchange** on the AVTools ribbon (Pro, and in the trial). See the Room Exchange User Guide.

Q: How do I lay out speakers in a room that doesn't exist in the architectural model?
Use **Add Temp Room** on the Rooms tab. Pick a number, name, and ceiling height, then trace the perimeter in your active plan view. AVTools creates a synthetic Temp Room row that behaves like a normal room for Configuration, Coverage, Circuiting, and Cabling. Temp rooms are excluded from Room Acoustics (placement scaffolds only). Once you place speakers, AVTools auto-resolves the underlying real Revit room (if any) on the next Rescan and re-stamps the speakers to that real room.

Q: How do I lay out speakers in a Revit room that has Area = 0 (unbounded)?
Use **Add Temp Room** on the Rooms tab — it's the unified workflow for both unbounded Revit rooms and spaces with no Revit Room at all. Select the row, trace the perimeter. AVTools stamps a Filled Region with the room number encoded in Comments; the next Rescan picks that polygon up as the room's layout boundary. The Filled Region is an annotation and does not propagate through Revit links.

Q: I placed speakers in a Temp Room but the underlying corridor still shows Not Started.
Click Rescan Rooms. AVTools runs the temp-to-real association pass on every Rescan — speakers in a Temp Room polygon that overlaps a real Revit room get re-stamped to the real room, and the real room's design state inherits the temp's. If the corridor still doesn't update, the temp polygon may be crossing more than one real room (watch for a warning dialog) or it may sit in genuinely unbounded space with no real room beneath.

Q: Why does my rack scan find 200 racks when I only have 6?
The keyword scan matches "rack" / "cabinet" / "enclosure" / Middle Atlantic product codes across every family in host + linked models. Furniture, IT racks, and network cabinets can light up. Use the Rack Family Selection dialog that appears after **Rack Locations**: tick only the families that should count as AV racks. Your choice persists — the dialog won't re-prompt for already-confirmed or already-rejected families. New families appearing in later scans flag with a "New" badge so they can be reviewed without disrupting existing selections.

Q: What loudspeaker data file formats are supported?
`.spk` (EASE SPK with optional sibling `.lob` / `.phs` / `.fed` / `.frd` / `.fvt`) and `.clf` (Common Loudspeaker Format, CLF1 + CLF2). Other formats — including `.gll`, manufacturer proprietary binary formats, and `.cf2` as balloon data — are not supported. If a manufacturer publishes `.gll` only, request CLF data from them.

Q: Does it calculate RT60 and STI?
Yes (Standard and Pro). Three RT60 formulas — Sabine, Norris-Eyring, Arau-Puchades — across eight octave bands (63 Hz to 8 kHz). STI is estimated from RT60, background noise, speaker Q, and N factor. Critical distance and D/R ratio are computed alongside.

Q: How is this different from D-Tools or EASE?
D-Tools is a system integration / proposal / project-management platform — strong on procurement and BOMs, not on Revit-native distributed-loudspeaker layout or acoustic analysis. EASE is a full acoustic

simulation suite — strong on auralization and ray tracing, not on Revit integration or circuiting/cabling. AVTools is BIM-native, focused on the AV consultant designing inside the architect's Revit model, and covers placement → coverage → RT60/STI → circuiting → cabling → coordination as one continuous workflow; Room Exchange hands a room to EASE when it needs full simulation.

Q: How do I get started?
Download the 14-day free trial from the products page; it unlocks every Pro feature, Room Exchange and SpecTool included. The free v1.2.1 (Revit 2022–2024) is available as a permanent free download for the core layout and coverage workflow. Both run as a dockable panel inside Revit; no external dependencies.

Q: Where do I report bugs or request features?
The About tab includes Bundle Logs (ZIP), Open Logs Folder, and Copy Support Info to gather diagnostic information. For a Room Exchange export, use Zip Support Files on its result dialog. Email support@bimacoustics.net with the bundled logs and a description of the issue or feature request.

Q: What was in the earlier 2.2 releases?
v2.2 brought Add Temp Room, the unified linked-file scan with a Source column, the Rack Family Selection dialog, Apply to Selected / Apply to All, in-color coverage maps on every view, and a set of reliability fixes. v2.2.1 added the MSI installer, the in-app EULA, the per-family loudspeaker filter dialog, user-level filter choices and cloud-delivered Lucius knowledge. v2.2.2 signed the installer and every DLL as J. Stevens BIM Acoustics LLC.
